

□

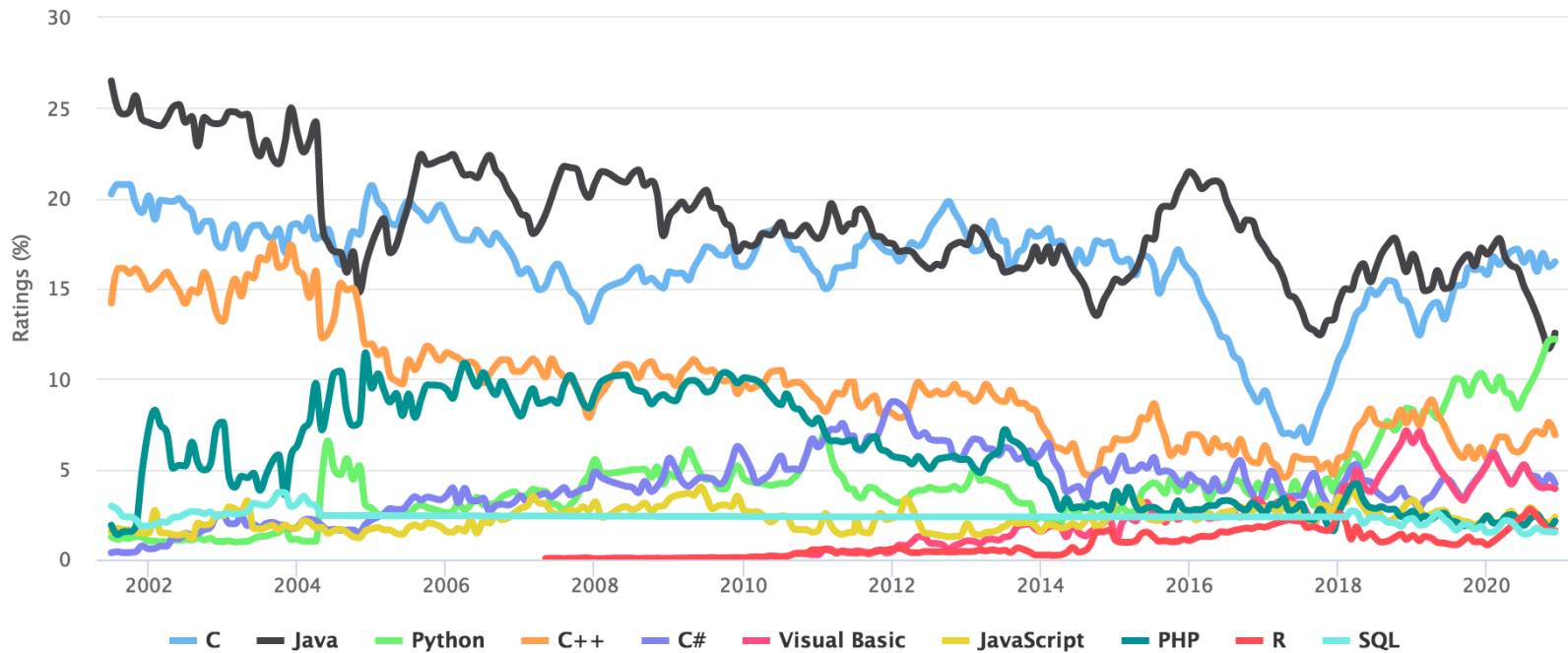
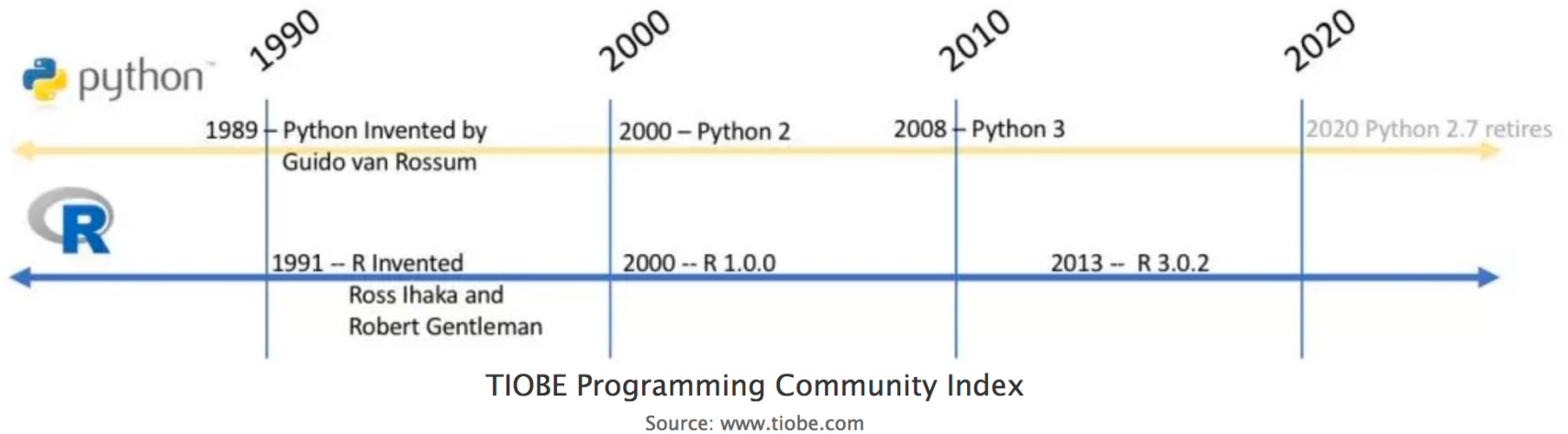
数据科学概述

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西安交通大学智能决策与机器学习中心
(Email: yao.s.wang@gmail.com)

2024. 5

Python与R的对比



Python与R的对比

- R主要用于学术与研究，越是接近统计学与数据科学，你就会越偏爱R；Python主要用于程序员想进入数据分析领域、掌握统计技能，工作环境越接近工程环境，你会越偏爱Python。
- R的社区生态主要是研究员，数据科学家、统计学家和量化研究员，而Python主要是程序员和开发者。
- 相比R，Python更简单易学且计算效率更高。但R的数据可视化效果远远强于Python。
- 如果从事纯统计学研究，用好R足够；如果从事深度学习研究或开发，用好Python足够。其他的数据科学家通常是“左手R右手Python”。

Python与R的对比



Brian Ray

@brianray

Follow



应读者要求，我要写一篇“在数据科学领域 Python vs R 语言”的文章。那么，大家希望我提到哪些方面呢？当然，这并不是什么信仰之战，这两个语言都非常好用，也同样都受到大部分的数据科学家支持，这也包括了

@ProjectJupyter

11:23 PM - 13 Apr 2018

13 Retweets 49 Likes



16



13



49



Hadley Wickham @hadleywickham · Apr 14



回复 @brianray @ProjectJupyter

比如把“vs”改成“和”？



3



17



159

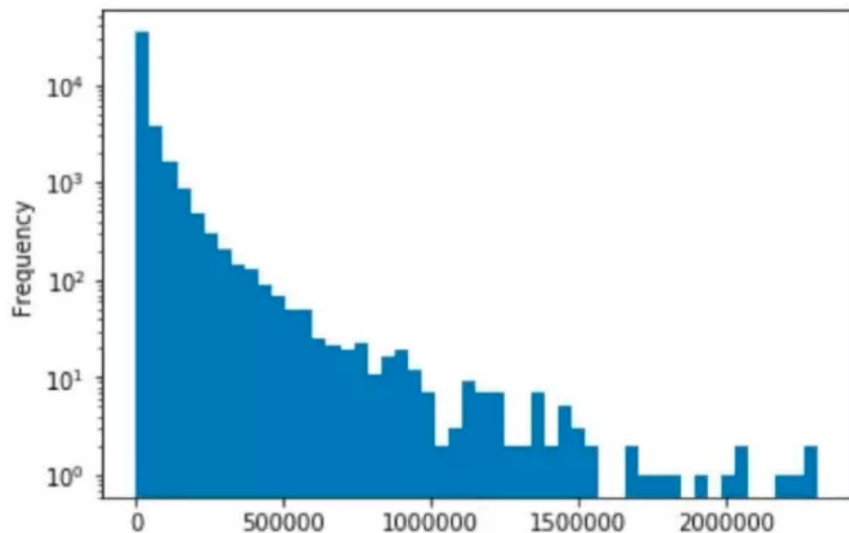


Python与R的对比

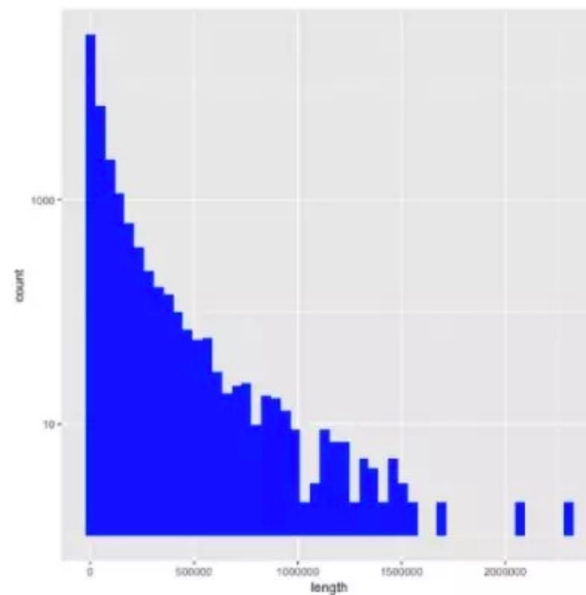
	Python	R	
将数据读取到数据表(DataFrame)中	9.7s	1min 11s	
列出特殊的基因序列ID	64ms	39ms	
列出特殊的形状	77ms	31ms	
分组计算记录数据的数量	284ms	271ms	
拆分表 ('GRCh38')	184ms	827ms	
对 sources 字段取样	750ms	655ms	
对 attributes 字段取样	183ms	1min 55s	<- !!?
应用正则表达式	177ms	1.3s	
绘制可视化图形	249ms	1s	

Python 在大部分的任务中计算优势明显

Python与R的对比



Python + Matplotlib



R + ggplot

R的可视化效果更好

Python的主要优点

- 设计严格：可读性强，易于维护
- 库很丰富：能够广泛应用于各种问题的场景
- 免费开源：具有高度可移植性，可在各类平台顺利工作
- 可扩展性：能够对R、C/C++等多种语言进行调用
- 可嵌入性：能够被集成到C/C++中，从而给程序用户提供脚本功能

Anaconda套件

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Individual Edition

Your data science toolkit

With over 20 million users worldwide, the open-source Individual Edition (Distribution) is the easiest way to perform Python/R data science and machine learning on a single machine. Developed for solo practitioners, it is the toolkit that equips you to work with thousands of open-source packages and libraries.

[Download](#)

Anaconda套件是一个大包的集合，里面预装了Conda(包管理器)、最新版本的Python、众多packages以及其他科学计算工具

<https://www.anaconda.com/products/individual>

Anaconda套件

Anaconda Installers

Windows 

Python 3.8

64-Bit Graphical Installer (457 MB)

32-Bit Graphical Installer (403 MB)

MacOS 

Python 3.8

64-Bit Graphical Installer (435 MB)

64-Bit Command Line Installer (428 MB)

Linux 

Python 3.8

64-Bit (x86) Installer (529 MB)

64-Bit (Power8 and Power9) Installer (279 MB)

<https://www.anaconda.com/products/individual>

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Applications on base (root) Channels



Datalore

Online Data Analysis Tool with smart coding assistance by JetBrains. Edit and run your Python notebooks in the cloud and share them with your team.

Launch



IBM Watson Studio Cloud

IBM Watson Studio Cloud provides you the tools to analyze and visualize data, to cleanse and shape data, to create and train machine learning models. Prepare data and build models, using open source data science tools or visual modeling.

Launch



JupyterLab

2.2.6

An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.

Launch



Notebook

6.1.4

Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.

Launch



PyCharm Professional

2020.3.1

A full-fledged IDE by JetBrains for both Scientific and Web Python development. Supports HTML, JS, and SQL.

Launch



Qt Console

4.7.7

PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.

Launch



Spyder

4.1.5

Scientific PYTHON Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features

Launch



Glueviz

1.0.0

Multidimensional data visualization across files. Explore relationships within and among related datasets.

Install



Orange 3

3.26.0

Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.

Install



RStudio

1.1.456

A set of integrated tools designed to help you be more productive with R. Includes R essentials and notebooks.

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pythonProject

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Search Packages

Name	Description	Version
✓ _ipyw_jlab_nb_ex...	A configuration metapackage for enabling anaconda-bundled jupyter extensions	0.1.0
✓ alabaster	Configurable, python 2+3 compatible sphinx theme.	0.7.12
✓ anaconda	Simplifies package management and deployment of anaconda	2020.11
✓ anaconda-client	Anaconda.org command line client library	1.7.2
✓ anaconda-project	Tool for encapsulating, running, and reproducing data science projects	0.8.4
✓ applaunchservices		0.2.1
✓ appnope	Disable app nap on os x 10.9	0.1.0
✓ appscript	Control applescriptable applications from python.	1.1.1
✓ argh	The natural cli.	0.26.2

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pandas

Name	Description	Version
✓ pandas	High-performance, easy-to-use data structures and data analysis tools.	1.1.3

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 Conda & Conda-Build Read	 Jupyter documentation Read	 Spyder documentation Read	 VSCode (python) Read	 Orange documentation Read	 Scikit-Learn Documentation Read	 PyTorch Documentation Read

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 Conda & Conda-Build Read	 Jupyter documentation Read	 Spyder documentation Read	 VSCode (python) Read	 Orange documentation Read	 Scikit-Learn Documentation Read	 PyTorch Documentation Read

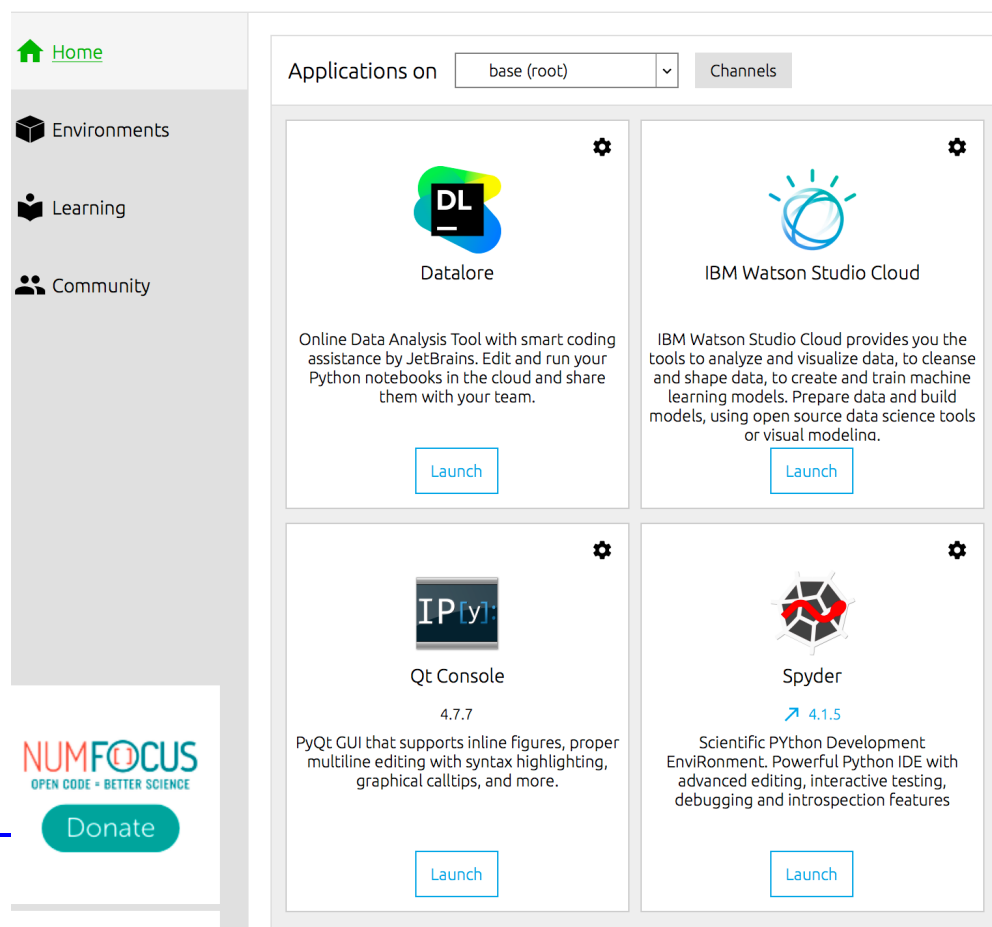
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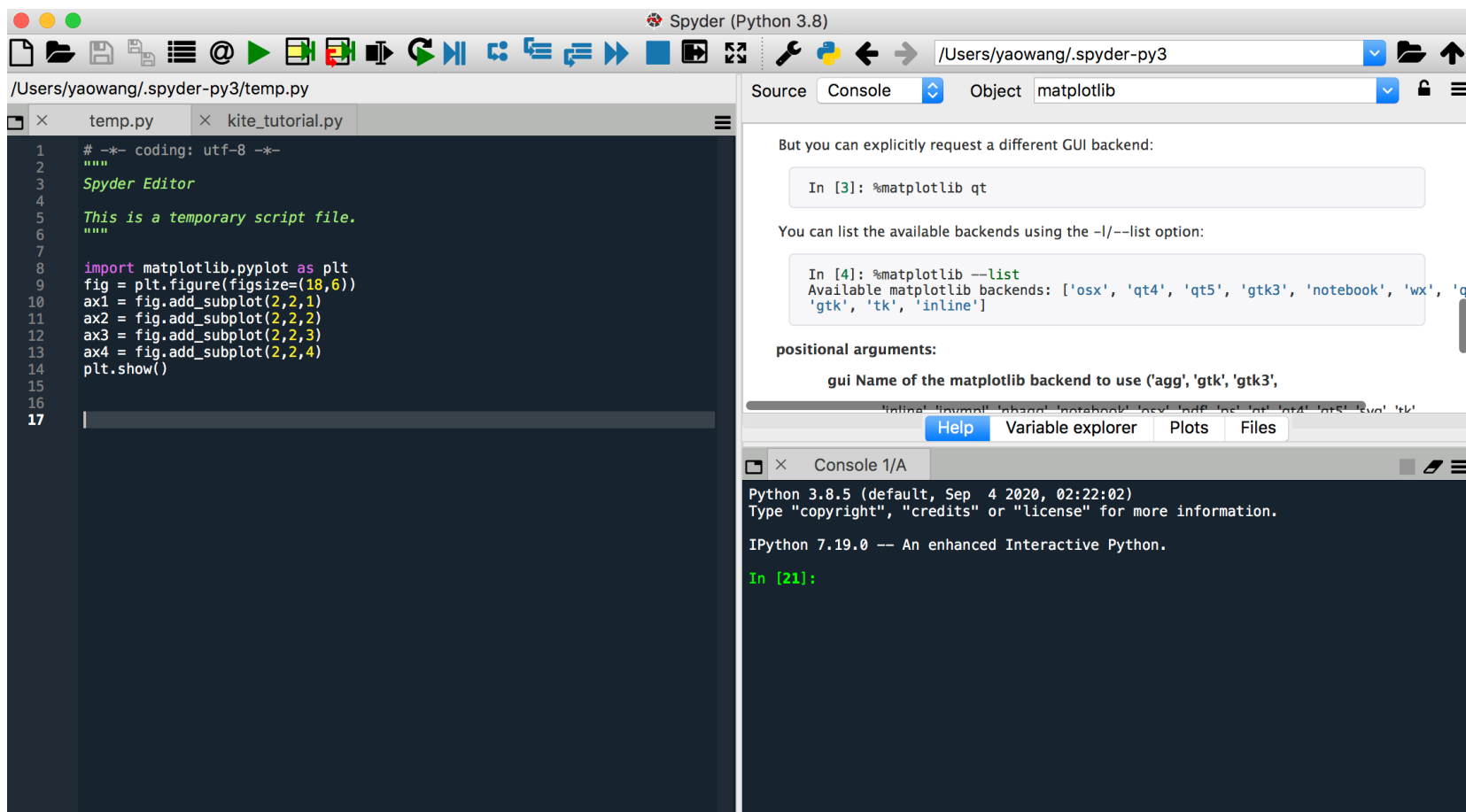
Documentation

Spyder介绍

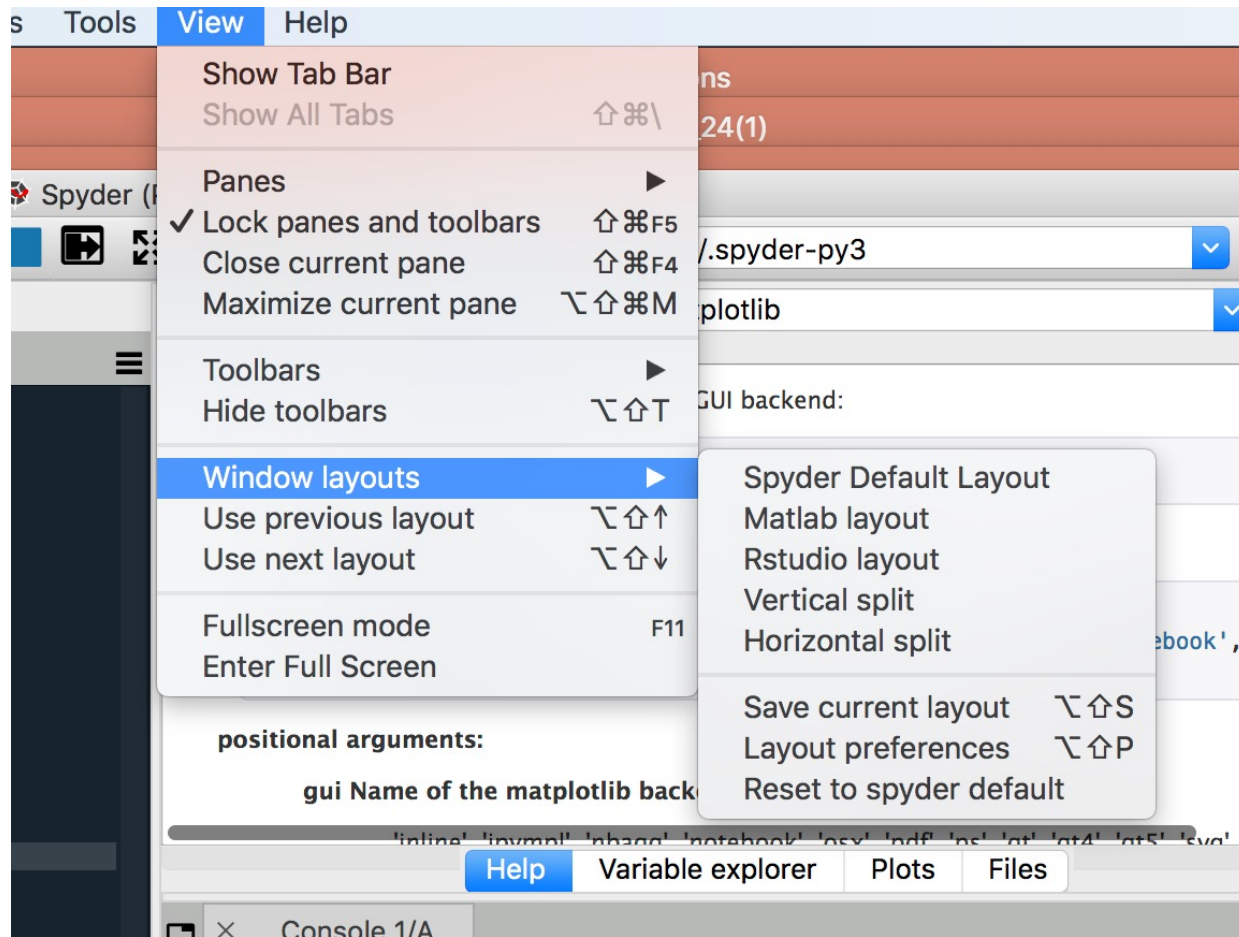
Spyder是使用Python语言进行科学计算的IDE。它结合了综合开发工具的高级编辑、分析、调试功能以及多种可视化功能，作用类似于R语言中的Rstudio。Anaconda可直接启动Spyder。



Spyder介绍



Spyder介绍



Spyder介绍

The screenshot displays the Spyder Python IDE interface. The top toolbar contains various icons for file operations, running, and debugging. The main window is divided into three panes:

- Script Editor:** Shows a Python script in `temp.py` with the following code:

```
1  # -*- coding: utf-8 -*-
2  """
3  Spyder Editor
4
5  This is a temporary script file.
6  """
7
8  import matplotlib.pyplot as plt
9  fig = plt.figure(figsize=(18,6))
10 ax1 = fig.add_subplot(2,2,1)
11 ax2 = fig.add_subplot(2,2,2)
12 ax3 = fig.add_subplot(2,2,3)
13 ax4 = fig.add_subplot(2,2,4)
14 plt.show()
15
16
17
```
- Console:** Shows the output of the script, including the Python version (3.8.5) and the IPython version (7.19.0). The prompt `In [21]:` is visible.
- Variable Explorer:** Displays a table of variables in the current namespace:

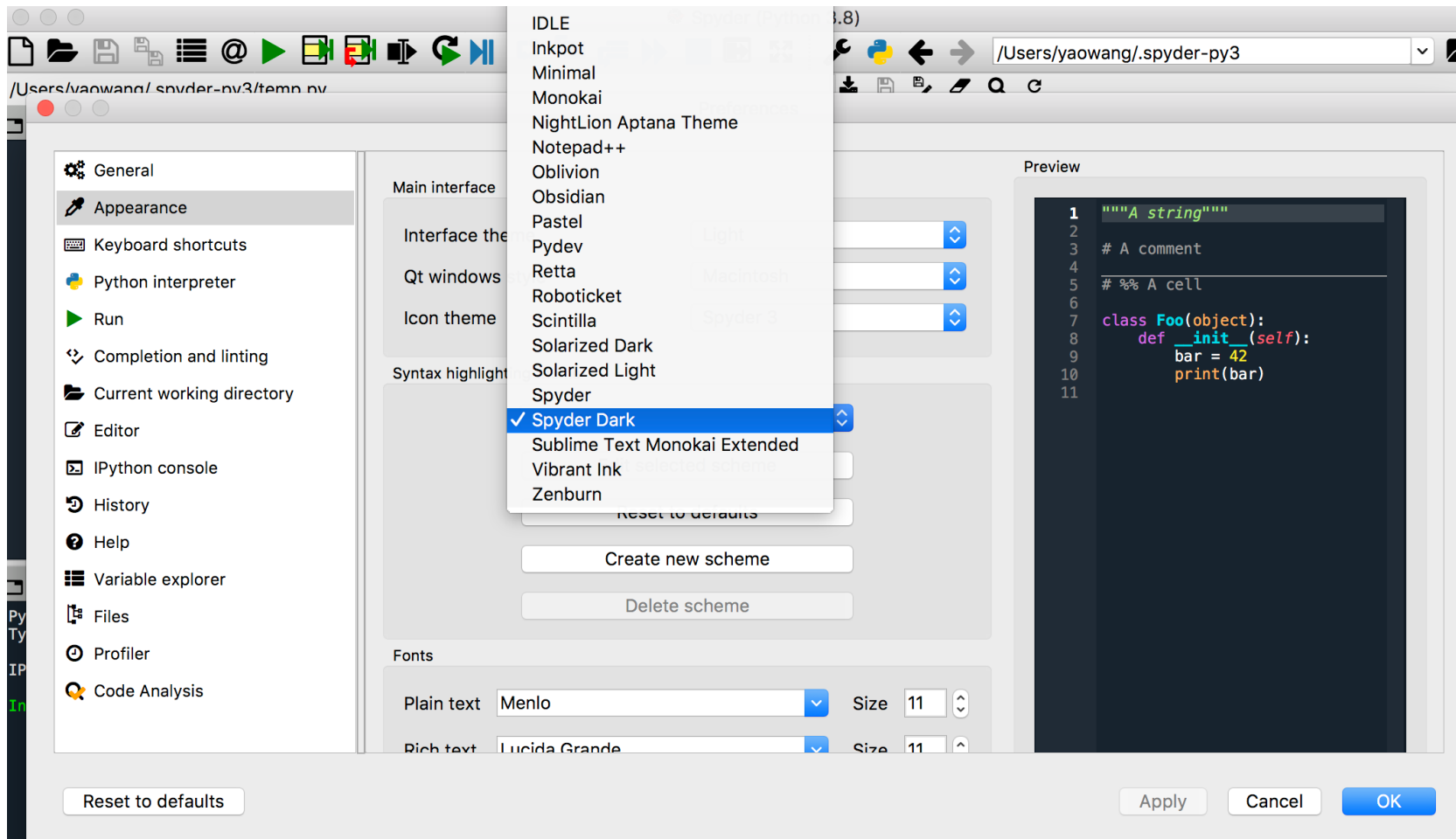
Name	Type	Size	Value
ax1	axes._subplots.AxesSubplot	1	AxesSubplot object of...
ax2	axes._subplots.AxesSubplot	1	AxesSubplot object of...
ax3	axes._subplots.AxesSubplot	1	AxesSubplot object of...
ax4	axes._subplots.AxesSubplot	1	AxesSubplot object of...
df	DataFrame	(150, 5)	Column names: length,...
fig	figure.Figure	1	Figure object of matp...
iris	utils.Bunch	7	Bunch object of sklearn.utils module
x	int	1	43

Below the variable explorer, the **Variable explorer** tab is active, showing the **Source** as `matplotlib` and the **Object** as `matplotlib`. The console output includes the following text:

```
Python 3.8.5 (default, Sep 4 2020, 02:22:02)
Type "copyright", "credits" or "license" for more information.
IPython 7.19.0 -- An enhanced Interactive Python.
In [21]:
```

The bottom status bar shows the following information: LSP Python: ready, Kite: indexing, conda: base (Python 3.8.5), Line 17, Col 1, UTF-8, LF, RW, Mem 68%.

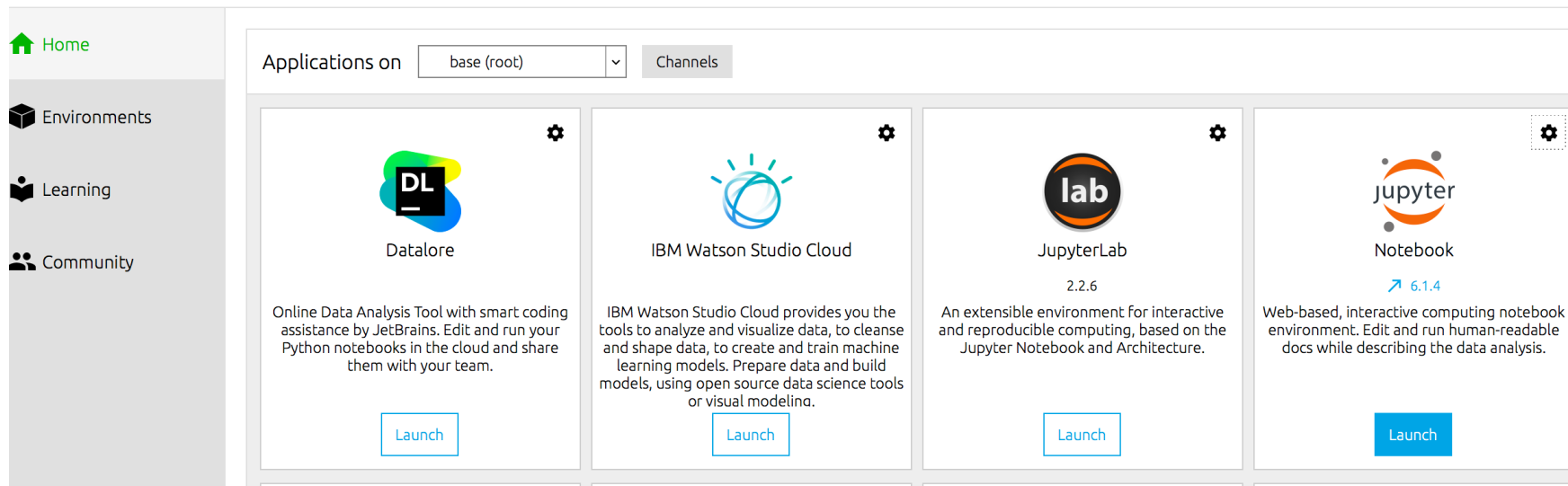
Spyder介绍



Spyder的优点

- 类Matlab/Rstudio设计，操作界面友好
- 资源丰富且查找便利，具有变量自动补全、函数调用提示以及随时随地访问文档帮助的功能
- 对初学者友好，即在其菜单栏中的【help】中给新用户提供了交互式的使用教程以及快捷方式的备忘单
- 工具丰富，功能强大，除一般IDE具有的组件以外，它还具有对象查看器、变量查看器、交互式命令窗口、交互式命令串口和历史命令窗口等组件

Jupyter Notebook介绍



“程序猿”里的“散文家”

Jupyter Notebook介绍

localhost:8890/tree

...

☆

 jupyter

Quit

Logout

FilesRunningClusters

Select items to perform actions on them.

0

/

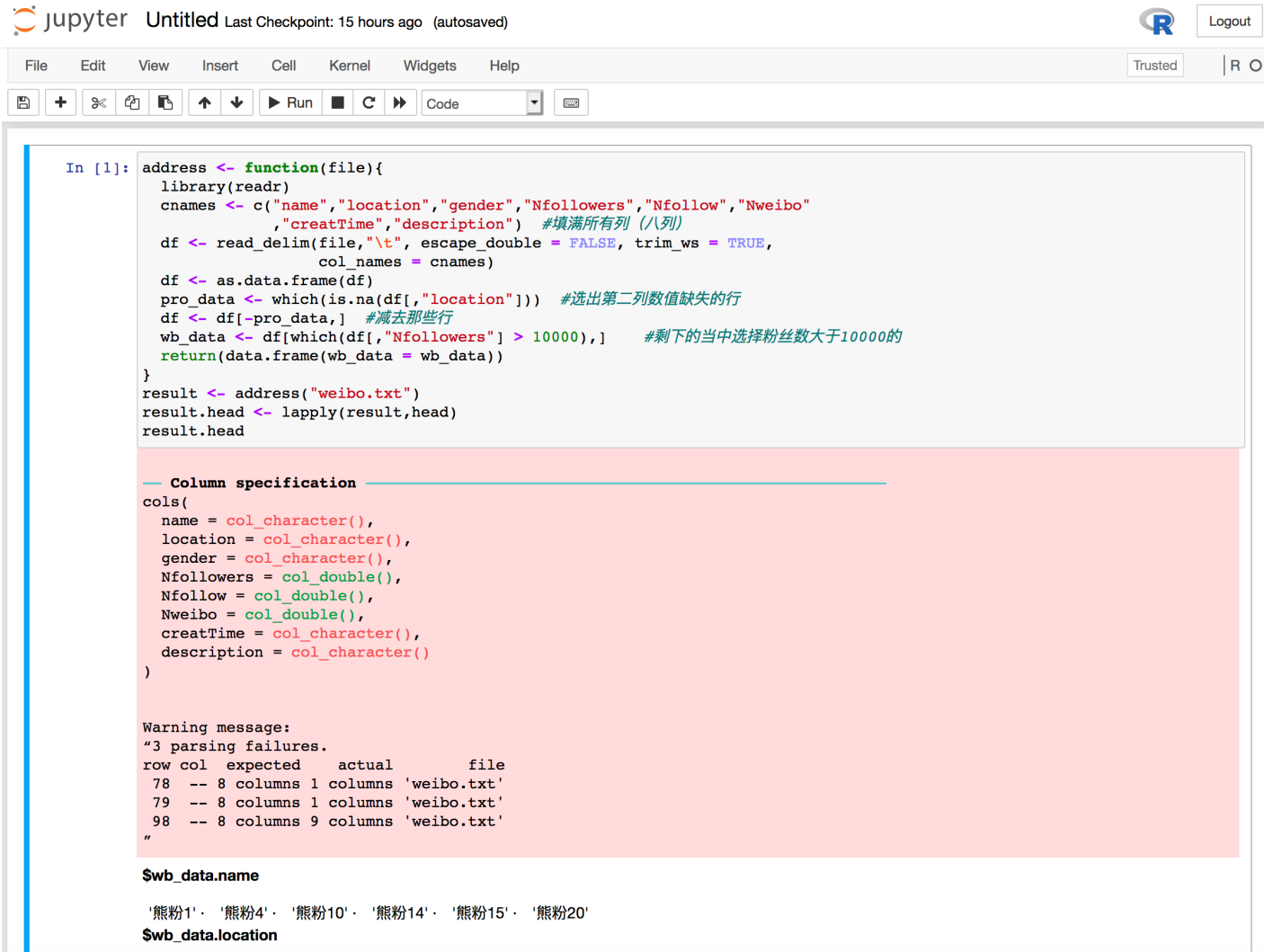
Name

Upload

New

☐				
☐		BizConf		
☐		MATLAB		
☐		TencentMeeting		
☐		Untitled.ipynb	Run	kB
☐		1400115281_report_pb.dump		
☐		Untitled2.pdf		16 hours ago 83.8 kB
☐		weibo.txt		2 years ago 13.1 kB
☐		招生介绍.docx		6 months ago 13.5 kB
☐		请假申请.docx		2 years ago 12.5 kB
☐		附件.docx		2 years ago 12.4 kB

Jupyter Notebook介绍



The screenshot displays a Jupyter Notebook interface. At the top, the title bar shows 'jupyter' with its logo, followed by 'Untitled' and 'Last Checkpoint: 15 hours ago (autosaved)'. On the right, there are 'Logout' and 'Trusted' buttons. Below the title bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', and 'Help'. A toolbar contains icons for saving, adding, deleting, and running code, along with a 'Run' button and a dropdown menu currently set to 'Code'. The main area shows a code cell with R code. The code defines a function 'address' that reads a file, filters rows based on missing 'location' values and a minimum number of followers (10000), and returns a data frame. Below the code, there is a section titled 'Column specification' which defines the expected data types for each column. A warning message follows, indicating 3 parsing failures in the file 'weibo.txt'. The output of the code shows the names and locations of the data rows.

```
In [1]: address <- function(file){
  library(readr)
  cnames <- c("name", "location", "gender", "Nfollowers", "Nfollow", "Nweibo",
    "createTime", "description") #填满所有列 (八列)
  df <- read_delim(file, "\t", escape_double = FALSE, trim_ws = TRUE,
    col_names = cnames)
  df <- as.data.frame(df)
  pro_data <- which(is.na(df[, "location"])) #选出第二列数值缺失的行
  df <- df[-pro_data,] #减去那些行
  wb_data <- df[which(df[, "Nfollowers"] > 10000),] #剩下的当中选择粉丝数大于10000的
  return(data.frame(wb_data = wb_data))
}
result <- address("weibo.txt")
result.head <- lapply(result, head)
result.head
```

— Column specification —

```
cols(
  name = col_character(),
  location = col_character(),
  gender = col_character(),
  Nfollowers = col_double(),
  Nfollow = col_double(),
  Nweibo = col_double(),
  createTime = col_character(),
  description = col_character()
)
```

Warning message:
"3 parsing failures.
row col expected actual file
78 -- 8 columns 1 columns 'weibo.txt'
79 -- 8 columns 1 columns 'weibo.txt'
98 -- 8 columns 9 columns 'weibo.txt'
"

\$wb_data.name

'熊粉1' '熊粉4' '熊粉10' '熊粉14' '熊粉15' '熊粉20'

\$wb_data.location

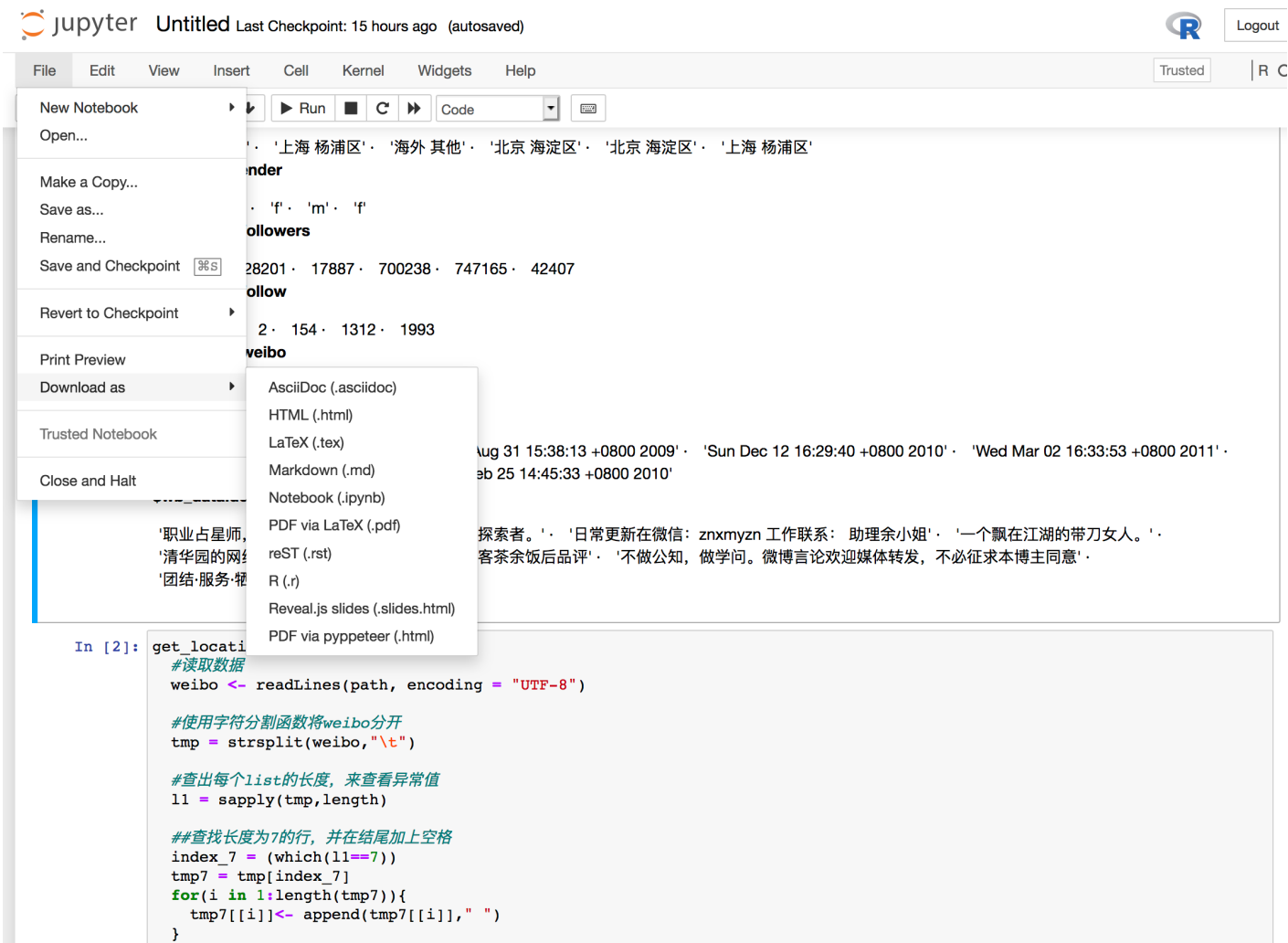
Jupyter Notebook介绍

```
In [2]: get_location <- function(path){  
  #读取数据  
  weibo <- readLines(path, encoding = "UTF-8")  
  
  #使用字符分割函数将weibo分开  
  tmp = strsplit(weibo, "\t")  
  
  #查出每个list的长度, 来查看异常值  
  l1 = sapply(tmp, length)  
  
  ##查找长度为7的行, 并在结尾加上空格  
  index_7 = (which(l1==7))  
  tmp7 = tmp[index_7]  
  for(i in 1:length(tmp7)){  
    tmp7[[i]] <- append(tmp7[[i]], " ")  
  }  
  
  #查找长度为8的行  
  index_8 = (which(l1==8))  
  tmp8 = tmp[index_8]  
  
  #将长度为7和8的行进行整合  
  weibo1 = c(tmp7, tmp8)  
  
  #将weibo1转化成dataframe  
  weibo2 = data.frame(matrix(unlist(weibo1), nrow=100, byrow=T),  
                           stringsAsFactors=FALSE)  
  
  #添加列名  
  names(weibo2) = c('name', 'location', 'gender', 'Nfollowers', 'Nfollow', 'Nweibo',  
                    'createTime', 'description')  
  
  #找出粉丝数大于10000的用户index  
  locations_index = which (as.numeric(weibo2$Nfollowers)>10000)  
  
  #返回找出对应index用户的所在地  
  return (list(weibo2[locations_index,]$location, weibo2))  
}
```

```
In [3]: p = "~/Downloads/weibo.txt"  
get_location(p)
```

```
1. '北京 朝阳区' · '上海 杨浦区' · '海外 其他' · '北京 海淀区' · '北京 海淀区' · '上海 杨浦区' · '北京 东城区' · '北京 海淀区' · '上海' ·  
   '海外 英国' · '北京 海淀区' · '重庆 渝中区' · '上海 徐汇区' · '北京 朝阳区' · '上海' · '北京' · '北京' · '北京 海淀区' · '北京 朝阳区' ·
```

Jupyter Notebook介绍



The screenshot shows the Jupyter Notebook interface. The top bar displays the Jupyter logo, the name 'Untitled', and the last checkpoint time 'Last Checkpoint: 15 hours ago (autosaved)'. The right side has a 'Logout' button and a 'Trusted' status indicator. The main menu bar includes 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', and 'Help'. The 'File' menu is open, showing options like 'New Notebook', 'Open...', 'Make a Copy...', 'Save as...', 'Rename...', 'Save and Checkpoint', 'Revert to Checkpoint', 'Print Preview', 'Download as', 'Trusted Notebook', and 'Close and Halt'. The 'Download as' sub-menu is open, listing various file formats: AsciiDoc (.asciidoc), HTML (.html), LaTeX (.tex), Markdown (.md), Notebook (.ipynb), PDF via LaTeX (.pdf), reST (.rst), R (.r), Reveal.js slides (.slides.html), and PDF via pypeteer (.html). The 'Markdown (.md)' option is highlighted. In the background, a code cell is visible with the following R code:

```
In [2]: get_location
#读取数据
weibo <- readLines(path, encoding = "UTF-8")

#使用字符分割函数将weibo分开
tmp = strsplit(weibo, "\t")

#查出每个list的长度, 来查看异常值
l1 = sapply(tmp, length)

##查找长度为7的行, 并在结尾加上空格
index_7 = (which(l1==7))
tmp7 = tmp[index_7]
for(i in 1:length(tmp7)){
  tmp7[[i]] <- append(tmp7[[i]], " ")
}
```

有些时候我们需要转成.md文件格式并进行后续处理

什么是Markdown?

- Markdown是一种可以使用普通文本编辑器编写的标记语言，通过简单的标记语法，它可以使普通文本内容具有一定的格式
- 它本质包含两个部分，一部分是纯文本的格式化语言，另一部分是将文本格式转换成其他形式(如PDF、HTML)的软件工具，即Markdown编辑器
- 它有着书写过程流畅、格式不随编辑器而改变、书写错误易发现等优点
- Markdown具有一系列衍生版本，用于扩展Markdown的功能（如表格、图像、数学公式、内嵌HTML等）

Markdown与互联网之子

豆瓣电影

搜索电影、电视剧、综艺、影人

影讯&购票 选电影 电视剧 排行榜 分类 影评 2019年度榜单 2019书影音报告

互联网之子 The Internet's Own Boy: The Story of Aaron Swartz



导演: 布莱恩·耐本伯格
编剧: 布莱恩·耐本伯格
主演: 亚伦·斯沃茨 / 蒂姆·伯纳斯-李 / 科利·多克托罗 / 彼得·艾克斯莱 / 劳伦斯·莱斯格 / 更多...
类型: 纪录片 / 传记
官方网站: www.takepart.com/internets-own-boy
制片国家/地区: 美国
语言: 英语
上映日期: 2014-01-20(圣丹斯电影节) / 2014-06-27(美国)
片长: 105分钟
又名: 斯沃茨的故事(台) / 网路之子: 亚伦·史沃兹 / 互联网之子: 亚伦·斯沃茨的故事
IMDb链接: [tt3268458](https://www.imdb.com/title/tt3268458)

豆瓣评分

9.1  12597人评价

5星  64.0%
4星  28.3%
3星  6.8%
2星  0.7%
1星  0.2%

好于 86% 纪录片

豆瓣

Aaron Swartz与John Gruber于2004年共同创建了Markdown语言

Markdown基本语法

编辑器最基本的功能，就是对文字本身加以处理。例如对文字加粗，在Markdown 中通过**** ****来实现。其实在 Word 中就对对应工具栏中的「字体」选项，同类的标记字符还有 ** ** 来实现斜体。

****演示粗体****

演示斜体

演示粗体

演示斜体

Markdown基本语法

相较于对「字」的编辑，更高一层的就是对「段落」的编辑，对应 Word 中其实也是工具栏的「段落」选项。和第一类字符稍有不同，这些字符会把一些段落变成特殊格式的段落。例如 + 实现列表，# 实现标题效果，> 将一段文字变为引用。

```
+ 演示列表
    + 列表还可以有层级

> 这是引用文字的效果
```

- 演示列表
 - 列表还可以有层级

这是引用文字的效果

Markdown基本语法

相较于对「字」的编辑，更高一层的就是对「段落」的编辑，对应 在 Word 中其实也是工具栏的「段落」选项。和第一类字符稍有不同，这些字符会把一些段落变成特殊格式的段落。例如 + 实现列表，# 实现标题效果，> 将一段文字变为引用。

```
# A First Level Header
```

```
## R Markdown
```

```
### 三级标题
```

A First Level Header

R Markdown

三级标题

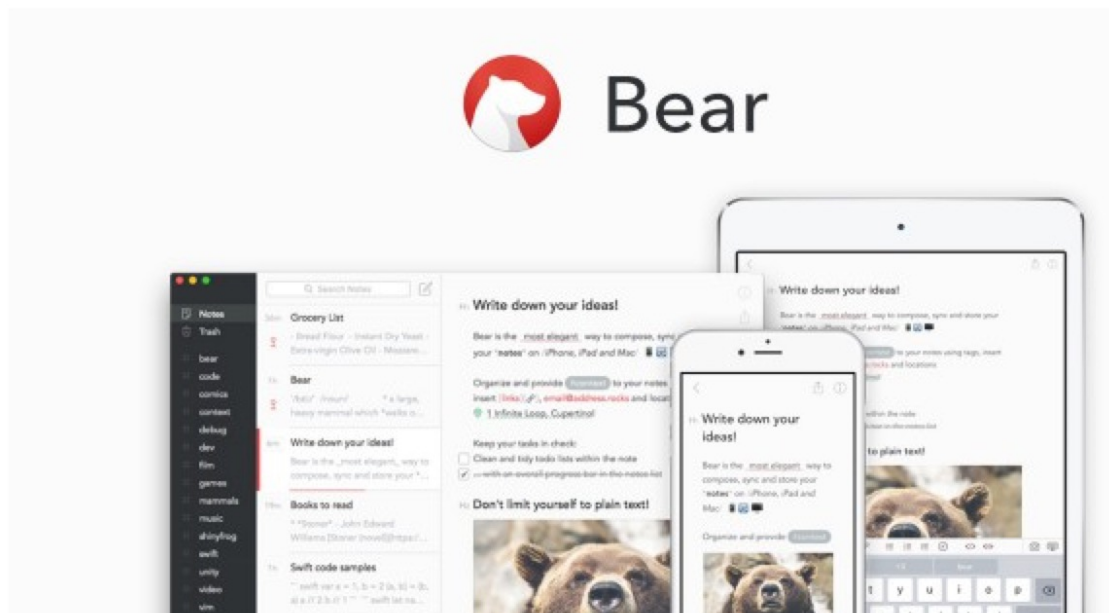
Markdown基本语法

正如 Word 中的「插入」选项一样，Markdown 也不仅仅只编辑文字，而是可以将不同的元素放入文档中。最常见的就是通过 `[]()` 来插入链接和 `![]()` 来插入图片（可以是本地图片也可以是网络图片）

[少数派](https://sspai.com)

)

少数派



Markdown语法在线学习

声明：这份文档派生(fork)于[繁体中文版](#)，在此基础上进行了繁体转简体工作，并进行了适当的润色。此文档用 Markdown 语法编写，你可以到这里[查看它的源文件](#)。「繁体中文版的原始文件可以[查看这里](#)」--By @riku

注：本项目托管于 [GitCafe](#)上，请通过"派生"和"合并请求"来帮忙改进本项目。

Markdown: Basics（快速入门） / (点击查看完整语法说明)

Getting the Gist of Markdown's Formatting Syntax

本页提供了 Markdown 的简单概念，[语法说明](#) 页提供了完整详细的文档，说明了每项功能。但是 Markdown 其实很简单就可以上手，此页文档提供了一些范例，并且每个范例都会提供输出的 HTML 结果。

其实直接试试看也是一个很不错的方法，[Dingus](#) 是一个网页应用程序，你可以把自己编写的 Markdown 文档转成 XHTML。

段落、标题、区块代码

一个段落是由一个以上的连接的行句组成，而一个以上的空行则会划分出不同的段落（空行的定义是显示上看起来像是空行，就被视为空行，例如有一行只有空白和 tab，那该行也会被视为空行），一般的段落不需要用空白或换行缩进。

Markdown 支持两种标题的语法，[Setext](#) 和 [atx](#) 形式。Setext 形式是用底线的形式，利用 `=`（最高阶标题）和 `-`（第二阶标题），Atx 形式在行首插入 1 到 6 个 `#`，对应到标题 1 到 6 阶。

区块引用则使用 email 形式的 `>` 角括号。

Markdown 语法:

Markdown语法在线学习

<< [访问 Wow!Ubuntu](#)

NOTE: This is Simplified Chinese Edition Document of Markdown Syntax. If you are seeking for English Edition Document. Please refer to [Markdown: Syntax](#).

声明: 这份文档派生(fork)于[繁体中文版](#), 在此基础上进行了繁体转简体工作, 并进行了适当的润色。此文档用 Markdown 语法编写, 你可以到这里[查看它的源文件](#)。「繁体中文版的原始文件可以[查看这里](#)。」--By [@riku](#)

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Markdown 语法说明 (简体中文版) / ([点击查看快速入门](#))

- [概述](#)
 - [宗旨](#)
 - [兼容 HTML](#)
 - [特殊字符自动转换](#)
- [区块元素](#)
 - [段落和换行](#)
 - [标题](#)
 - [区块引用](#)
 - [列表](#)
 - [代码区块](#)
 - [分隔线](#)
- [区段元素](#)
 - [链接](#)
 - [强调](#)
 - [代码](#)
 - [图片](#)
- [其它](#)
 - [反斜杠](#)
 - [自动链接](#)
- [感谢](#)
- [Markdown 免费编辑器](#)

<http://www.wowubuntu.com/markdown/index.html>

Markdown文件利器--Typora

Typora

Home

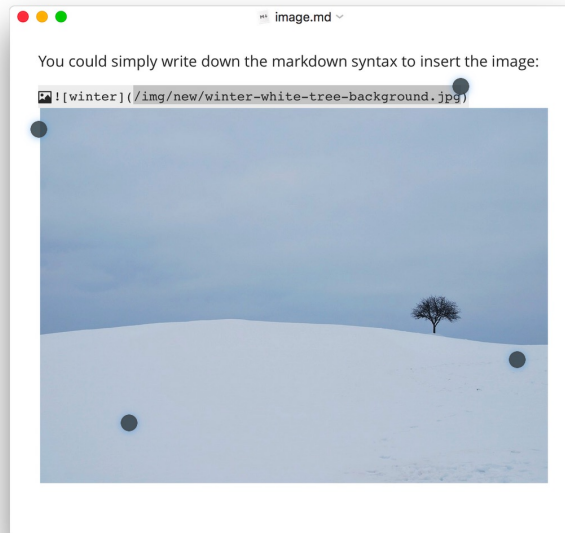
Features

Download

Themes

Simple, yet Powerful

Images, Headers, Lists, Tables, Code Fences, Mathematics, Diagrams, Inline Styles, etc..



/* DISPLAY IMAGES, RATHER THAN TEXT FOR IMAGES WHEN WRITING */

Markdown文件利器--Typora



Untitled(1).md

```
address <- function(file){  
  library(readr)  
  cnames <-  
  c("name", "location", "gender", "Nfollowers", "Nfollow", "Nweibo"  
    , "createTime", "description") #填满所有列 (八列)  
  df <- read_delim(file, "\t", escape_double = FALSE, trim_ws =  
  TRUE,  
    col_names = cnames)  
  df <- as.data.frame(df)  
  pro_data <- which(is.na(df[, "location"])) #选出第二列数值缺失的行  
  df <- df[-pro_data,] #减去那些行  
  wb_data <- df[which(df[, "Nfollowers"] > 10000),] #剩下的当中选  
  择粉丝数大于10000的  
  return(data.frame(wb_data = wb_data))  
}  
result <- address("weibo.txt")  
result.head <- lapply(result, head)  
result.head
```

```
• [36m—• [39m • [1m• [1mColumn specification• [1m• [22m  
• [36m—• [39m  
cols(
```

Markdown文件利器--Typora

\$wb_data.name

1. '熊粉1'
2. '熊粉4'
3. '熊粉10'
4. '熊粉14'
5. '熊粉15'
6. '熊粉20'

\$wb_data.location

HTML </>

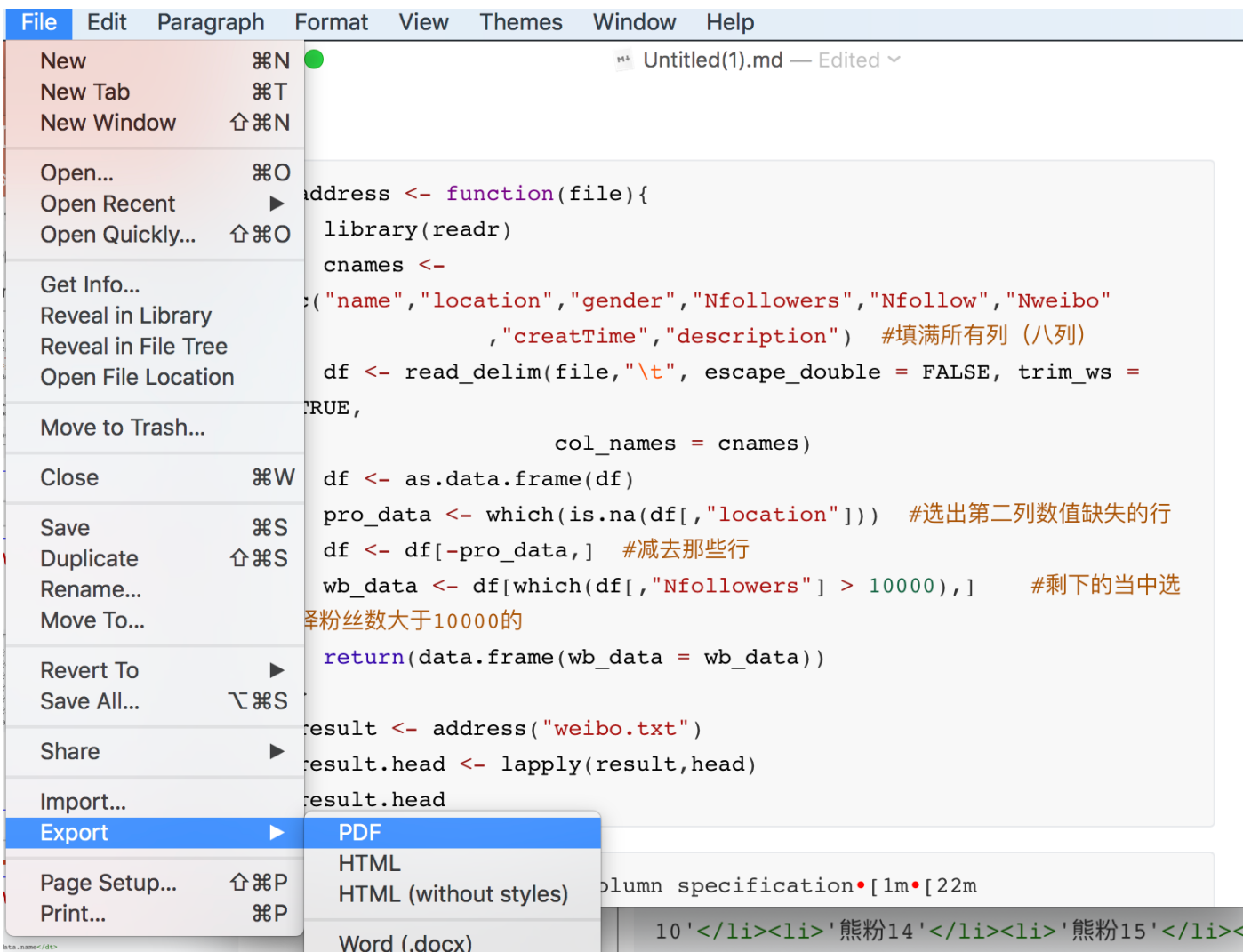
Edit

Markdown文件利器--Typora

HTML ✓

```
<dl>
  <dt>$wb_data.name</dt>
  <dd><style>
.list-inline {list-style: none; margin:0; padding: 0}
.list-inline>li {display: inline-block}
.list-inline>li:not(:last-child)::after {content: "\00b7";
padding: 0 .5ex}
</style>
<ol class=list-inline><li>'熊粉1'</li><li>'熊粉4'</li><li>'熊粉
10'</li><li>'熊粉14'</li><li>'熊粉15'</li><li>'熊粉20'</li></ol>
</dd>
  <dt>$wb_data.location</dt>
  <dd><style>
.list-inline {list-style: none; margin:0; padding: 0}
.list-inline>li {display: inline-block}
.list-inline>li:not(:last-child)::after {content: "\00b7";
padding: 0 .5ex}
</style>
```


Markdown文件利器--Typora



Markdown文件利器--Typora

Untitled(1).pdf ×		Untitled.pdf					
				5 / 10			
				125%			
熊粉90	北京 朝阳区	f	376	349	500	Fri Sep 11 20:48:40 +0800 2009	
熊粉1	北京 朝阳区	f	371068	183	2862	Wed Nov 24 15:54:44 +0800 2010	职业占星师，占星专栏作者，心理学、神秘学探索者。
熊粉4	上海 杨浦区	f	728201	202	11	Mon Aug 31 15:38:13 +0800 2009	日常更新在微信：znxmyzn 工作联系：助理余小姐
熊粉5	北京 东城区	f	508	336	536	Mon Mar 22 22:54:18 +0800 2010	愤青 宝妈 白领 普通青年 伪小资
熊粉8	浙江 杭州	m	2189	15	48	Mon Feb 06 11:02:23 +0800 2012	淘宝商城 tmall 天猫 数据挖掘 数据分析
:	:	:	:	:	:	:	:
熊粉65	上海	f	399	464	296	Sun May 09 13:02:23 +0800 2010	ceibs, hr service provider

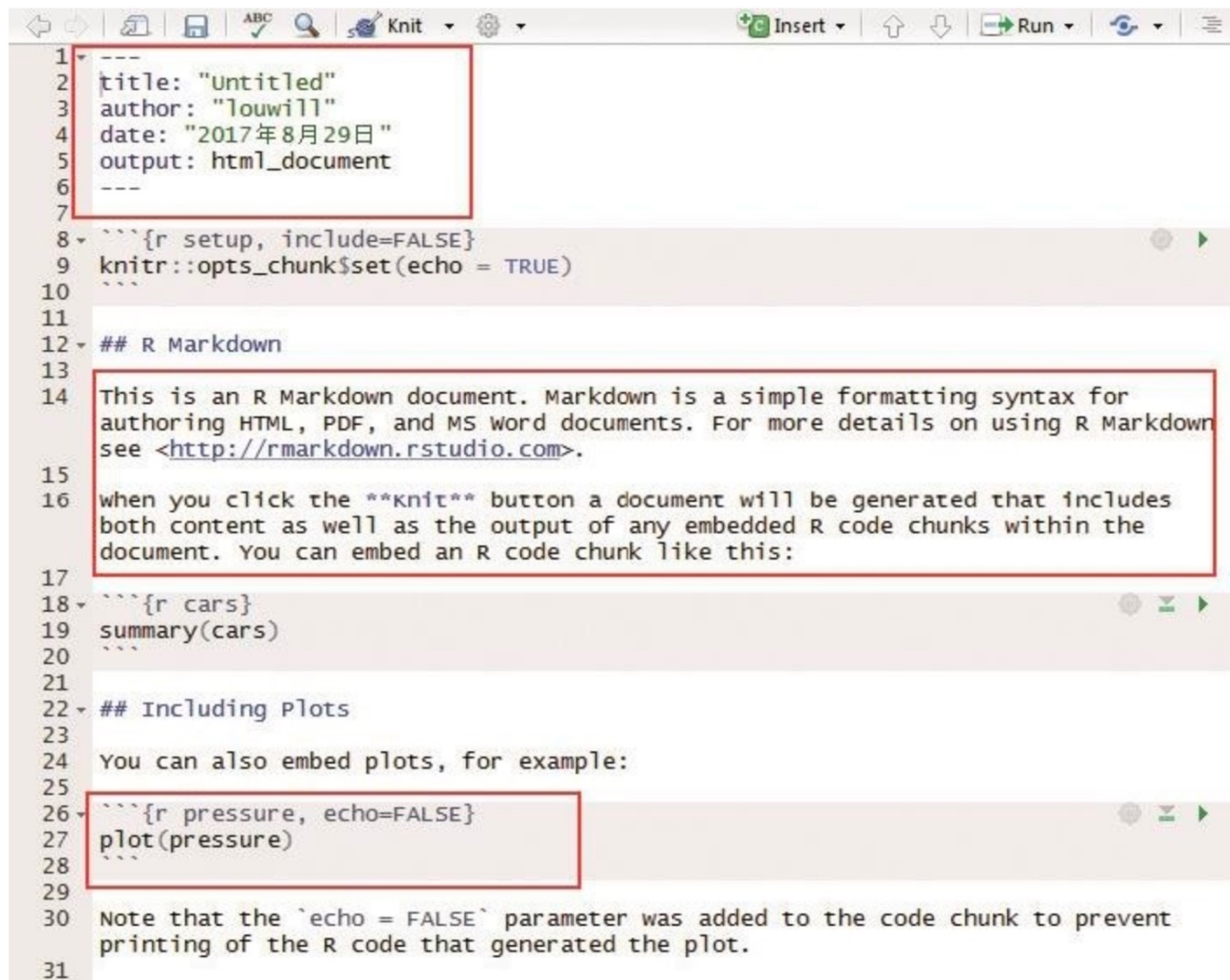
为什么要使用R Markdown?

- 使用Rmd最大的好处就是，可以轻松的重现你的数据分析工作（[make your work reproducible](#)）
- 从原始数据的读入、清洗到最后的分析挖掘过程，每一步的R code都被清晰地记录
- 阅读你的Rmd结果的人可以完整地审查你的分析思路 and 过程。
- Rmd可将数据分析代码和结果以文件的形式输出为相应的报告(如HTML、PDF等)

Rmd的文档结构

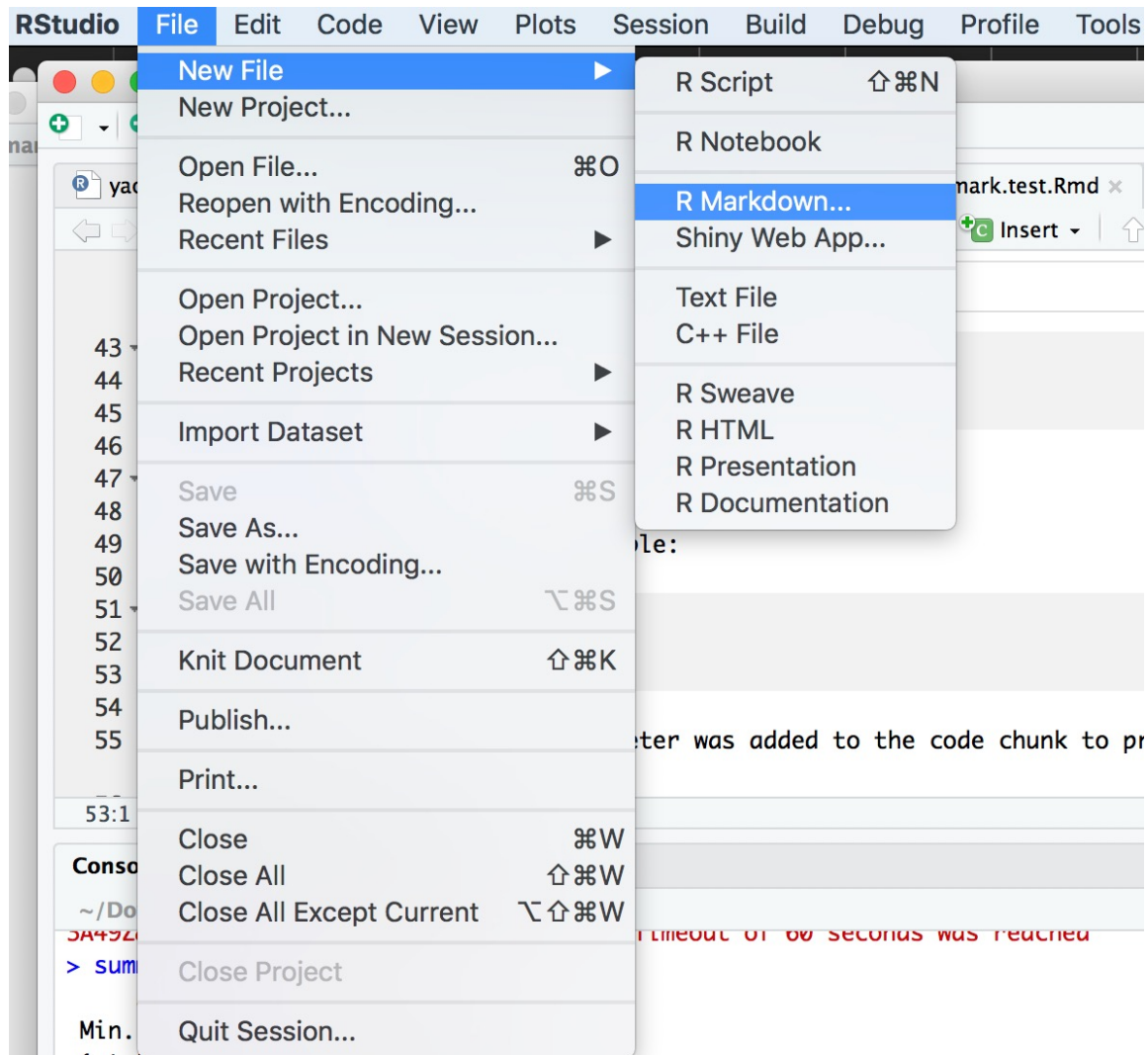
- 一份完整的Rmd文档通常由以下三部分构成：
 - `---`：符号内的YAML(yet another multicolumn layout)渲染参数
 - 可自由写作的text文本
 - `````：符号内的R代码块
- R就已经按照这三大内容配置好了模板文档，只需在此基础上按照个人需求对这三个结构进行参数设置和代码修改即可

Rmd的文档结构

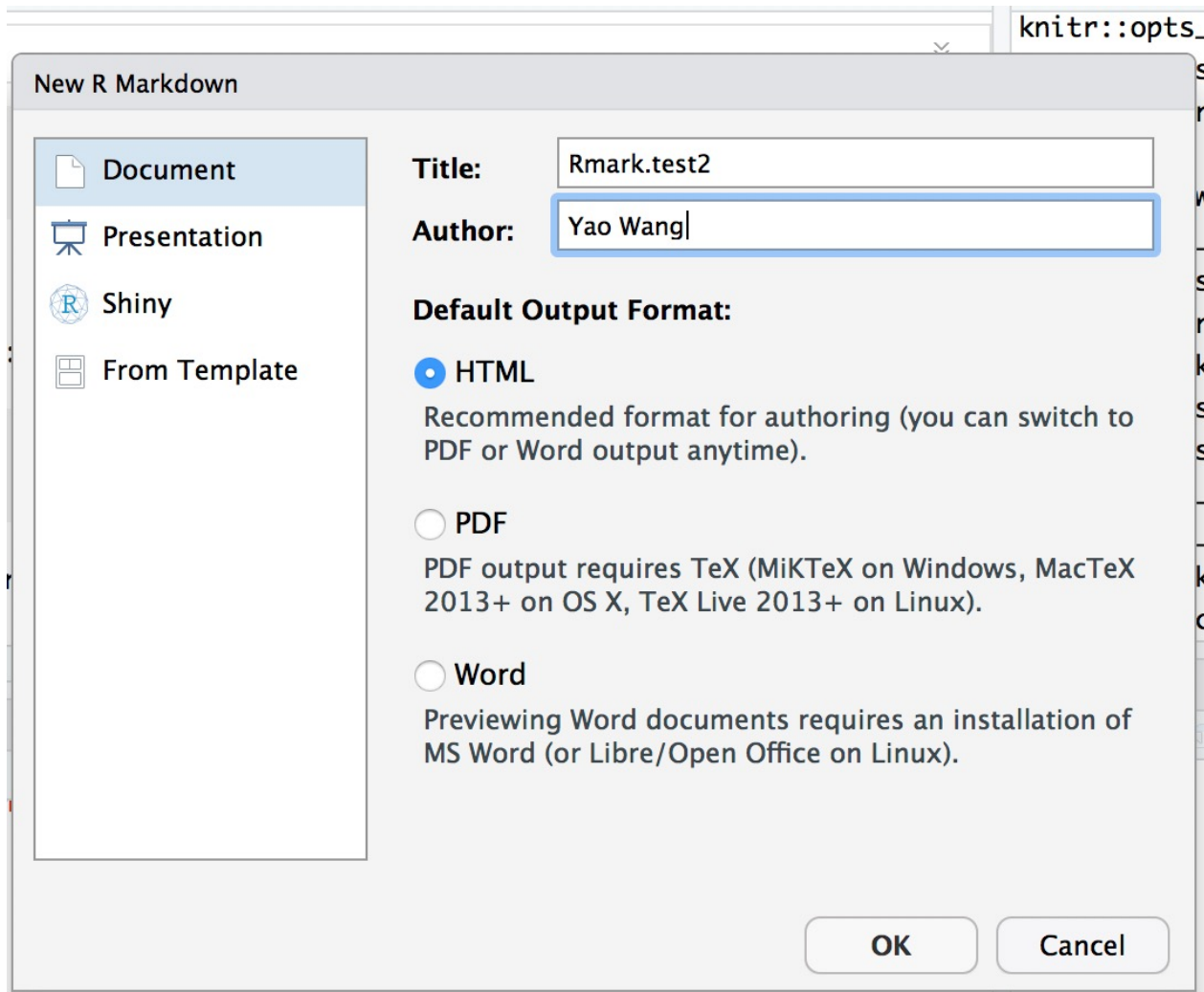


```
1 ---
2 title: "Untitled"
3 author: "louwill"
4 date: "2017年8月29日"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for
15 authoring HTML, PDF, and MS word documents. For more details on using R Markdown
16 see <http://rmarkdown.rstudio.com>.
17
18 when you click the **knit** button a document will be generated that includes
19 both content as well as the output of any embedded R code chunks within the
20 document. You can embed an R code chunk like this:
21
22 ```{r cars}
23 summary(cars)
24 ```
25
26 ## Including Plots
27
28 You can also embed plots, for example:
29
30 ```{r pressure, echo=FALSE}
31 plot(pressure)
32 ```
33
34 Note that the `echo = FALSE` parameter was added to the code chunk to prevent
35 printing of the R code that generated the plot.
```

Rmd的文档结构



Rmd的文档结构

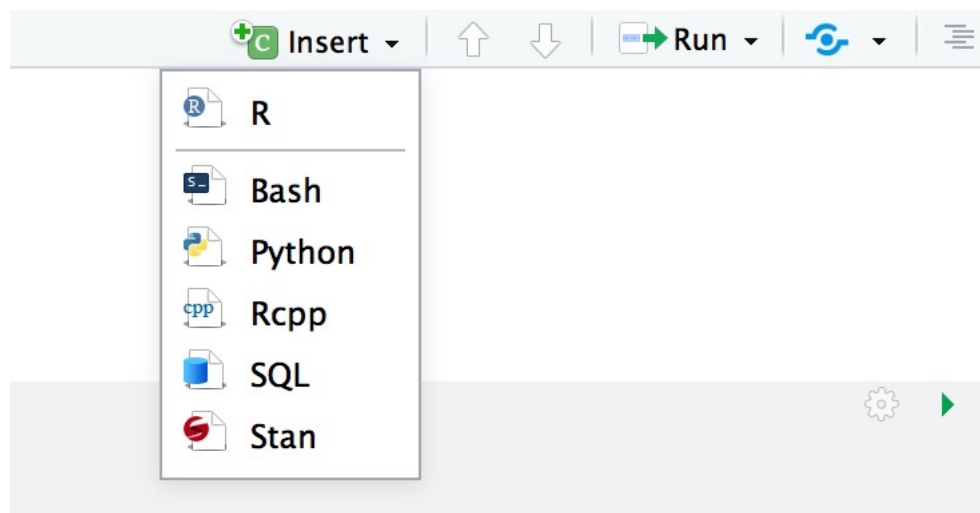


Rmd的文档结构

```
1 ---
2 title: "Rmark.test2"
3 author: "Yao Wang"
4 date: "12/18/2020"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF,
15 and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
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29 ```{r pressure, echo=FALSE}
30 plot(pressure)
31 ```
32
33 Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R
34 code that generated the plot.
```

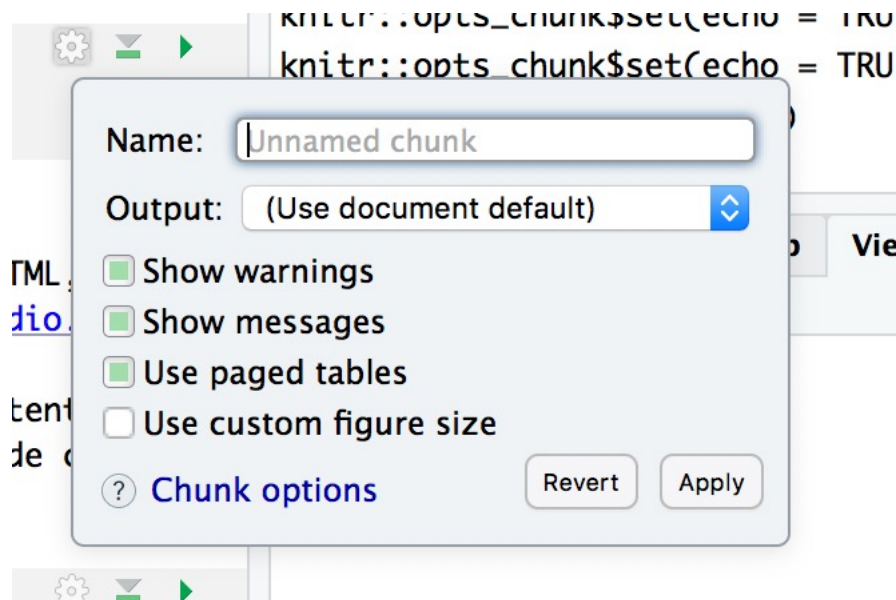

插入代码块

除了手动输入`` `符号插入代码之外，最方便的插入代码的方法莫过于通过菜单栏的insert来执行插入代码的功能了



插入代码块

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插入代码块

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```
`` `{r pressure, echo=FALSE}  
plot(pressure)  
`` `
```

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

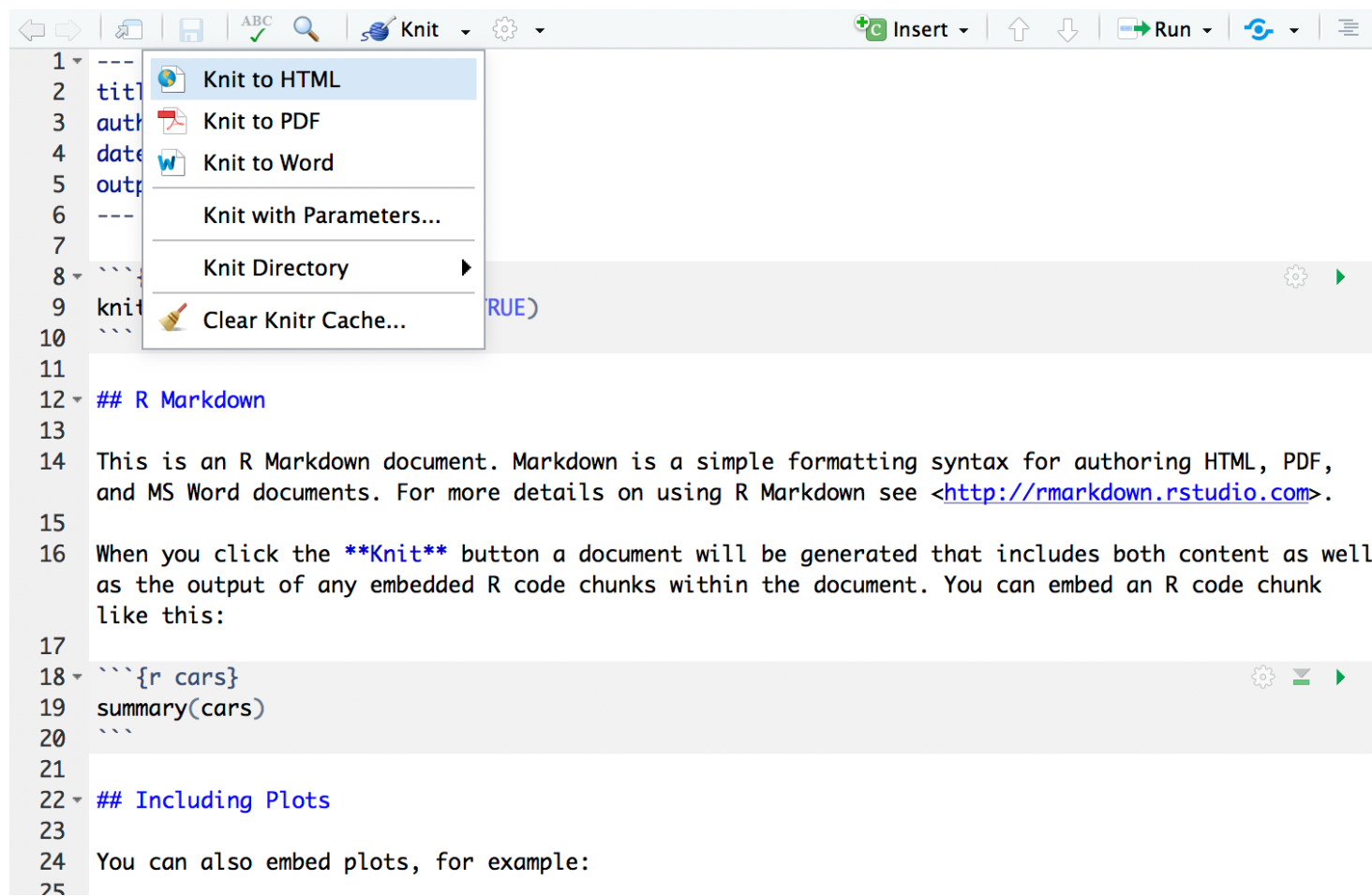
可以在代码块里手动输入命令参数进行设置

Rmd的输出



首先，R Markdown会将Rmd文件转变为一个名为knitr的文档，knitr可以理解为一个由纯文本和代码交织在一起的文档；然后再将这个文档转变为一个新的Markdown文件（.md）；最后由pandoc转变为任意指定的文档格式。pandoc是一种标记语言转换工具，可以实现不同标记语言之间的转换。

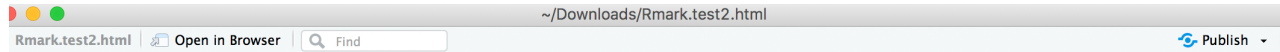
Rmd的输出



The screenshot shows the RStudio interface. At the top, the 'Knit' menu is open, displaying options: 'Knit to HTML' (selected), 'Knit to PDF', 'Knit to Word', 'Knit with Parameters...', 'Knit Directory', and 'Clear Knitr Cache...'. Below the menu, the R Markdown document is visible. It starts with a title 'R Markdown' and a description of the format. A code chunk is shown with the R code `summary(cars)`. The document also includes a section titled 'Including Plots'.

```
1 ---
2 title: R Markdown
3 author:
4 date:
5 output:
6 ---
7
8
9 knitr::knit()
10
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF,
15 and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
16
17 When you click the Knit button a document will be generated that includes both content as well
18 as the output of any embedded R code chunks within the document. You can embed an R code chunk
19 like this:
20
21
22 ```{r cars}
23 summary(cars)
24 ```
25
26 ## Including Plots
27
28 You can also embed plots, for example:
```

Rmd的输出



Rmark.test2

Yao Wang

12/18/2020

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed      dist
##  Min.   : 4.0   Min.   : 2.00
##  1st Qu.:12.0   1st Qu.: 26.00
##  Median :15.0   Median : 36.00
##  Mean   :15.4   Mean    : 42.98
##  3rd Qu.:19.0   3rd Qu.: 56.00
##  Max.   :25.0   Max.    :120.00
```

Including Plots

You can also embed plots, for example:



Rmd的其他输出

- 交互式文档 Interactive Documents
 - HTML Widgets
 - Shiny
- 仪表盘 Dashboards
 - Dashboard with HTML Widgets
 - Dashboard with Shiny
 - Dashboard with Storyboard
- 演示文稿 Presentations
 - Beamer slideshow
 - Slidy slideshow
 - reveal.js slideshow
- 书籍 Books
- 网站 Websites
- 模板 Templates
- R 包小品文 Package Vignettes

Rmd速查表

R Markdown

速查表

详情请查看 rmarkdown.rstudio.com

rmarkdown 0.2.50 更新: 8/14



由 supstat ANALYTICS 翻译

1. 工作流程 R Markdown是通过R来书写可重复动态报告的一种格式。利用它在幻灯片、html文档、Word文件中嵌入R代码和结果。报告制作流程:

i. 打开 - 打开扩展名为.Rmd的文件



ii. 编写 - 利用简易的R Markdown 语法编写内容



iii. 嵌入 - 嵌入R代码, 生成报告所需的输出结果

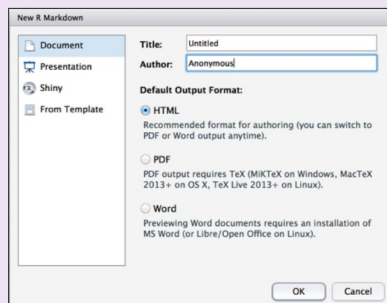


iv. 提交 - 用输出结果替代R代码, 把报告输出为幻灯片、pdf、html或微软Word文件。



2. 打开文件 以保存为扩展名.Rmd的文本文件开始, 或打开一个RStudio Rmd模板

- 点击菜单栏中的:
File ► New File ► R Markdown...
- 窗口弹出。选择.Rmd文件的输出类型
- 单击指定输出类型 (亦可随后修改)
- 点击确认



3. 简易标记 之后, 用纯文本书写你的报告。利用markdown语法来如何设计在最终报告中的文本格式。

语法

Plain text
End a line with two spaces to start a new paragraph.
italics and *_italics_*
****bold**** and **__bold__**
superscript^{2^}
~~~strikethrough~~~  
[\[link\]\(www.rstudio.com\)](http://www.rstudio.com)  
  
# Header 1  
## Header 2  
### Header 3  
#### Header 4  
##### Header 5

#### 显示

Plain text  
End a line with two spaces to start a new paragraph.  
*italics* and *italics*  
**bold** and **bold**  
superscript<sup>2</sup>  
~~strikethrough~~  
[link](#)  
  
**Header 1**  
**Header 2**  
**Header 3**  
**Header 4**

<https://rstudio.com/wp-content/uploads/2015/03/rmarkdown-chinese.pdf>



# Rmd如何导出中文PDF

■ 通过Rmd输出中文版pdf报告，要解决以下两个核心问题：

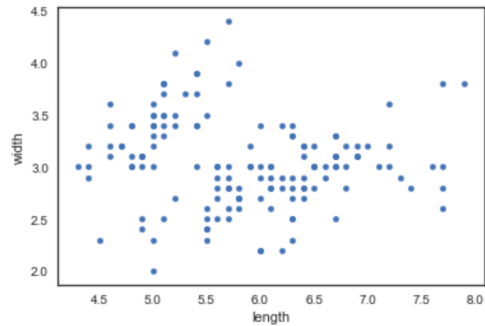
- 导出PDF需要配置TeX环境，以下两种方案都可以：
  - 安装TinyTex([参见Yihui Xie个人主页](#))
  - 安装任一LaTeX套件 (TeX Live\MiKTeX\MacTeX)
- PDF中文显示问题，下面两种方案中更推荐后者：
  - 在YAML中写入LaTeX的一些设置
  - 使用rticles包中的CTeX Documents模板

```
install.packages("rticles")
```

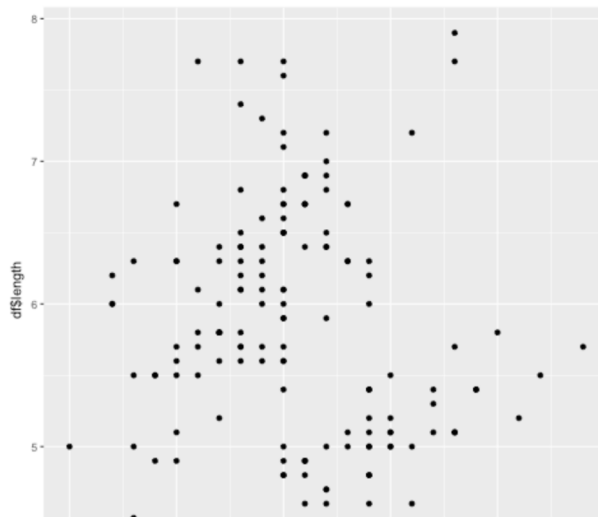
# Jupyter下的Python与R混编

```
In [3]: df.plot(kind="scatter", x="length", y="width")
```

```
Out[3]: <matplotlib.axes._subplots.AxesSubplot at 0x10cbf2898>
```



```
In [5]: %%R -i df
# may need to install.packages ggplot2
library('ggplot2')
ggplot(df, aes(x=df$width, y=df$length)) + geom_point()
```



## 在Jupyter中安装R内核



阿拉斯加肥肥q

且歌且行

8 人赞同了该文章



知乎 @詹大师

Jupyter Notebook（此前被称为 IPython notebook）是一个交互式笔记本，支持运行 40 多种编程语言。Jupyter Notebook 的本质是一个 Web 应用程序，便于创建和共享文学化程序文档，支持实时代码，数学方程，可视化和 markdown。用途包括：数据清理和转换，数值模拟，统计建模，机器学习等等。

<https://zhuanlan.zhihu.com/p/46799561>

## 2.三步实现 R内核的安装

安装过程很简单，你只需要跟着我一步一步地复制代码到R，运行即可。甚至可以连我的解释都不用看，直接把代码复制粘贴就ok。（ps：安装的过程中不要同时打开jupyter软件）

在R中安装镜像下载工具——devtools包 打开RStudio，输入以下代码：

```
install.packages('devtools') #在R中安装devtools库  
# 或者
```

利用镜像工具从github网站下载安装R kernel：

```
devtools::install_github('IRkernel/IRkernel')  
# 或者 devtools::install_local('IRkernel-master.tar.gz')
```

在R中注册激活R kernel

```
IRkernel::installspec() # to register the kernel in the current R installatio
```

**注意：** 在Mac下，上述命令需要通过Terminal操作

 yaowang — R — 80x24  
[(base) YaodeMacBook-Pro:~ yaowang\$ r]

```
R version 4.0.3 (2020-10-10) -- "Bunny-Wunnies Freak Out"  
Copyright (C) 2020 The R Foundation for Statistical Computing  
Platform: x86_64-apple-darwin17.0 (64-bit)
```

```
R is free software and comes with ABSOLUTELY NO WARRANTY.  
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.
```

```
Natural language support but running in an English locale
```

```
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.
```

```
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.
```

```
> install.packages('devtools')  
--- Please select a CRAN mirror for use in this session ---  
devtools::install_github('IRkernel/IRkernel')  
IRkernel::installspec()
```

# rpy2 - R in Python



Documentation Docker MyBinder Extensions

## rpy2 - R in Python



rpy2 is an interface to **R** running embedded in a **Python** process.

**pypi** **v3.3.6** **on Pypi**

```
yaowang — -bash — 80x24
Last login: Fri Dec 25 13:20:29 on ttys000
You have new mail.
(base) YaodeMacBook-Pro:~ yaowang$ pip install rpy2
Requirement already satisfied: rpy2 in /opt/anaconda3/lib/python3.8/site-packages (3.3.6)
Requirement already satisfied: pytest in /opt/anaconda3/lib/python3.8/site-packages (from rpy2) (0.0.0)
Requirement already satisfied: pytz in /opt/anaconda3/lib/python3.8/site-packages (from rpy2) (2020.1)
```

可以方便的在Python中使用  
R中流行的第三方包

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*Thank you for your  
attentions !*

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